

Ans 1) Explanation of Logic behind both the strategies

The overall strategy for Bitcoin (BTC) and Ethereum (ETH) is different from each other.

Bitcoin: -

In Bitcoin, the entry conditions for long and short trades are very strict but conditions are more relaxed for exiting a long and short trade. While working on the strategy, it has been noticed that long trades are executed more than short trades which shows that bitcoin is relatively a stable, upward moving cryptocurrency. On further restricting the conditions beyond a certain limit, the number of trades reduces significantly, which is not good for a robust strategy.

Ethereum: -

The entry conditions are a bit more relaxed for entering long and short trades but conditions are strict for exiting those trades. This strategy has emerged as the final one, while working on Ethereum. We arrived at this strategy because when the team was trying to create a robust strategy for Ethereum, we observed that price movements of Ethereum were very drastic, creating a very high volatility. In a high volatility period the signals may change rapidly. Even if we were applying strict conditions for entering the market (long/short), but the signals turned out to be either false or the signal changed in the course of taking trades. So, we decided that we will enter the market for taking long or short trades, but will check conditions thoroughly when can we exit the trade. Hence, strategy has strict exit conditions but moderate entry conditions. In Ethereum, it has been observed that more short trades are taken in comparison to long trades and the strategy is profitable in this way. It also tells that since market is very volatile, downtrend may occur more frequently and short selling will help to generate profit in this price movement scenario.

Detailed strategy for Bitcoin:

Long Entry:

When EMA20 is greater than EMA50, Heiken Ashi Open is less than EMA20, Heiken Ashi close is greater than EMA20, RSI smoothed is above 50, Hawkes close is greater than the closing price, Chaikin volatility is below 25, MACD is less than MACD signal line, KST is less than KST signal line, elder ray index confirms bullish trend and in Directional Movement index, +DI is greater than -DI and +DI is above 10, then a long buy signal is generated.

Long Exit:

When EMA20 is less than EMA50, Kalman filtered price is less than the closing price, Heiken Ashi open is greater than EMA20 and Heiken Ashi close is less than EMA20, then the long buy signal is closed, and the trade has been exited.

Short Entry:

When EMA6 is less than EMA10, Heiken Ashi Open is greater than EMA6, Heiken Asi Close is less than EMA6, MACD is greater than MACD signal line, Chebyshev filtered price is less than the closing price, RSI smoothed is below 40 and +DI is less than -DI, then a short trade is entered.

Short Exit:

When EMA6 is above EMA10, Heiken Asi open is less than EMA6, Heiken Ashi Close is above EMA6, percentage change indicator is less than or equal to the rolling threshold and Aroon oscillator is above -10, then the short signal is closed, and the trade has been exited.

Detailed Strategy for Ethereum:

Long Entry:

When EMA12 is greater than EMA26, Heiken Ashi Open is less than EMA20, Heiken Ashi close is above EMA20 and Chaikin Volatility is below 55 then then a long buy signal is generated.

Stricter Long Entry: (Triggered when holding time exceeds a threshold period)

In addition to Long Entry conditions, now the Chaikin Volatility should be below 50 and smoothed RSI should be above 60. Under these conditions, a stricter long entry trade will get started.

Long Exit:

When EMA12 is less than EMA26, Heiken Ashi Open is greater than EMA20, Heiken Ashi close is less than EMA20 and Chande Momentum Oscillator of the interval just before the interval where trade is executing is greater than or equal to 20, then the long buy signal is closed, and the trade has been exited.

Short Entry:

When EMA12 is less than EMA26, Chebyshev Filtered price is less than the closing price and directional movement index confirms a downtrend/bearish trend, then a short trade is entered.

Stricter Short Entry: (Triggered when holding time exceeds a threshold period)

In addition to the short entry conditions, if smoothed RSI is below 30, then a stricter short buy signal is generated.

Short Exit:

When commodity channel index is above -20, Aroon oscillator is above -50, Chande Momentum Oscillator of the interval just before the interval where trade is executing is less than or equal to 30, Donchian Middle of previous interval to the trade executing interval is less than closing price of current interval and percentage change indicator is less than or equal to rolling threshold, then the short signal is closed, and the trade has been exited.

Ans 2) Risk Management Techniques: -

Bitcoin:

During experimentation, we observed the presence of noise which could not be attributed to the underlying trends. If unfiltered, noise can lead to misinterpretation of market conditions, causing poor decision-making and increased risk. So we have used Kalman filters and powerful high pass filters used in signal processing like Chebyshev filters. We saw that Ethereum has a lot of volatility and needed a filter with a sharp cutoff. Hence, we preferred Chebyshev filters over Butterworth filters. We used these filters on both Bitcoin and Ethereum to achieve improved results.

Ethereum:

During experimentation, we observed that strategies that can capture underlying trends are working better for Ethereum. Our final strategy and signal generation logic for Ethereum had very high a time to recovery (TTR) of 664. We realized this happened because of our strict exit conditions. Some trades were simply not closing because of the strict exit conditions. We decided to cap the maximum holding time of a trade by implementing a 'time calculator' which keeps a track of the holding time for a trade. If holding time exceeds 100 days, then the trade is closed and re-entry into the trade is evaluated on stricter conditions than before. If the trade going on was long, then stricter long conditions are checked and similarly if the trade going on was short, then stricter short conditions are evaluated. We believe that both Bitcoin and Ethereum prices are affected by market sentiment which depends on the quarterly performance of many large companies closely linked to these cryptocurrencies. We also applied a modified Fourier transform in order to find out the most meaningful cycle under the threshold of 125 days. We kept a threshold of 125 days, because we did not want very long or macro cycles. On application of this transform, we got 93 days as the result. We rounded off the number to 100 days and have used that everywhere in order to avoid micro-optimizations.

Ans 3) Our Learnings

Our learning curve was quite steep in this experience. We initially applied deep learning, time series, bagging and boosting approaches. None of them seemed to work to our liking. Then we decided to come back to the basics and work on simple strategies. In the course of this competition, we understood that simple rule based strategies can be robust enough to handle the volatility and uncertainty in these markets. The most crucial learning point was that we understood how Ethereum is different from Bitcoin in terms of price movements and what works for Bitcoin may not work for Ethereum. We learnt that how trend-based indicators, mean aversion indicators and volatility based indicators are important in these types of market trends. Cryptocurrencies are traded across hundreds of exchanges globally. Each exchange has its own trading volume data, making it difficult to get a comprehensive and reliable measure of total trading volume. Hence, based on the team's observations, we concluded that volume-based indicators perform poorly for Bitcoin and Ethereum. Concepts from the field of signal processing also turned out to be useful in filtering noisy signals and capturing underlying trends. Hence, we understood that a robust trading strategy requires a logical, multi-faceted approach and a lot of patience.